

CY 202P : Organic Chemistry Laboratory

Course Code : CY 202P

Course Name : Organic Chemistry Laboratory

L-T-P-C : 0-0-4-2

Intended for : BS Chemical Sciences

Prerequisite : None

Mutual Exclusion: None

Approval: 42nd Senate

Course Contents

- Module I: **Basic Lab Techniques**

1. To determine type and detection of extra elements (N, S, Cl, Br, I) in organic compounds
2. Thin layer chromatography (TLC) and calculation of R_f values
3. Separation of 2 organic compounds by paper chromatography
4. Purification of organic compounds by crystallization.

- Module II: **Organic Qualitative Analysis**

1. Separation of two components from the binary mixture of organic compounds (Solid-Solid, Solid-Liquid).

- Module III: **Organic Preparations** (any 6)

1. Preparation of paracetamol
2. Preparation of aspirin
3. Preparation of phthalimide from phthalic anhydride
4. Preparation of 2:4-DNP derivative of aldehyde or ketone
5. Preparation 4-chloro benzyl alcohol from 4-chloro benzaldehyde
6. Base catalyzed Aldol condensation
7. Preparation 4-iodonitrobenzene from 4-nitroaniline by Sandmeyer Reaction
8. Preparation of Glucosazone derivative of Glucose
9. Preparation of quinone from hydroquinone
10. Preparation of Oxime derivative of Ketones.

Textbooks:

1. **Vogel's book of Practical Organic Chemistry**, 5th Edition, Longman Scientific & Technical, 2006.
2. Pavia, Lampman, Kriz & Engel, **Organic Chemistry A Lab Manual**, Cengage Learning, 2009.



References:

1. Frederick A. Bettelheim, **Experimental physical chemistry**, Saunders, 1971.
2. A. Ghosal, B. Mahapartra, A. K. Nad, **An Advanced Course in Practical Chemistry**, New Central Book Agency Pvt Ltd, 2000.